CLASS 10 BIOLOGY — CHAPTER: LIFE PROCESSES

Detailed Notes (NCERT / CBSE Board-Oriented)

1. INTRODUCTION

Life processes = the basic functions performed by living organisms to maintain their life and to keep them alive.

Criteria of life:

- **Visible movement** is NOT a definite criterion (e.g., a tree, a sleeping animal appear still but are alive).
- **Molecular movement** inside the body is the real indicator of life.
- Growth, respiration, reproduction, response to stimuli are additional criteria.

Why are life processes needed? Living structures are made of molecules that keep breaking down due to wear and tear. To repair and maintain them, energy is required. This energy comes from **food (nutrition)** and is released by **respiration**, transported by the **transportation system**, and waste produced is removed by **excretion**.

Four main life processes:

1. Nutrition
2. Respiration
3. Transportation
4. Excretion

> Unicellular organisms exchange materials directly with the environment by diffusion. Multicellular organisms have specialised tissues/organs because all cells are not in direct contact with the environment; diffusion alone is too slow and insufficient.

2. NUTRITION

Nutrition: The process of taking in food and utilising it for energy, growth, repair and maintenance.

Why needed: Source of energy + raw materials for building the body.

2.1 Modes of Nutrition

Mode	Definition	Examples
Autotrophic	Organism makes its own food from simple inorganic substances (CO_2, H_2O) using sunlight	Green plants, blue-green algae (cyanobacteria)
Heterotrophic	Organism depends on other organisms for food	Animals, fungi, most bacteria

Types of Heterotrophic Nutrition:

- **Saprophytic (Saprotrophic):** Feeds on dead and decaying organic matter by secreting digestive enzymes outside the body (extracellular digestion). → Fungi (bread mould, Rhizopus, yeast), mushrooms, many bacteria.

- **Parasitic:** Derives nutrition from a living host, harming it. → Cuscuta (amarbel), ticks, lice, leeches, tapeworm, Plasmodium.
- **Holozoic:** Takes in whole/complex food, then digests it internally. → Amoeba, Paramecium, humans.

2.2 AUTOTROPHIC NUTRITION — PHOTOSYNTHESIS

Definition: The process by which autotrophs take in CO_2 and water and convert them into carbohydrates in the presence of sunlight and chlorophyll.

Equation:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Raw materials: CO_2, water, sunlight, chlorophyll.

Main events of photosynthesis:

1. **Absorption of light energy by chlorophyll.**
2. **Conversion of light energy into chemical energy** and **splitting (photolysis) of water molecules** into hydrogen and oxygen.
3. **Reduction of CO_2 to carbohydrates.**

> These steps need not occur one after the other immediately. E.g., in desert plants (CAM plants) stomata open at night, CO_2 is taken in and stored as an acid, which is used during the day.

Site: Chloroplast (contains chlorophyll) — mostly in mesophyll cells of leaves. Chloroplast is called the **"kitchen of the cell."**

Entry of raw materials:

- **CO_2** → through **stomata** (tiny pores on leaf surface). Large amounts of gaseous exchange also occur across the surface of stems, roots and leaves.
- **Water** → absorbed from soil by roots → xylem → leaves.
- **Minerals** (N, P, Fe, Mg) → from soil as ions. **Nitrogen** is essential for proteins; taken as nitrates/nitrites or as organic nitrogen.

Stomata:

- Each stoma is bounded by two **guard cells**.
- **Opening:** Guard cells swell (turgid) when water flows in → pore opens.
- **Closing:** Guard cells shrink (flaccid) → pore closes.
- Stomata close mainly to **prevent excessive water loss by transpiration** when CO_2 is not required.

Experiments to remember:

- **Starch test (iodine test)** — proves starch is formed in photosynthesis. Destarch plant → keep in sunlight → boil leaf in water → boil in alcohol (to remove chlorophyll) → add iodine → **blue-black** colour shows starch.
- **Variegated leaf experiment** — proves chlorophyll is necessary (only green portions turn blue-black).
- **Experiment with KOH** (absorbs CO_2) — proves CO_2 is necessary.

2.3 HETEROTROPHIC NUTRITION EXAMPLES

Amoeba (holozoic, unicellular):

- Takes in food using temporary finger-like extensions — **pseudopodia**, which fuse over the food particle forming a **food vacuole**.
- Complex substances broken down inside the food vacuole by enzymes → diffuse into cytoplasm.
- Undigested material moved to the surface and thrown out (egestion).

Paramecium: Has a definite shape; food is taken in at a specific spot (**oral groove**) by the movement of **cilia**.

2.4 NUTRITION IN HUMAN BEINGS

Alimentary canal: Mouth → Oesophagus → Stomach → Small intestine → Large intestine → Rectum → Anus. **Associated glands:** Salivary glands, Liver, Pancreas.

(a) Mouth / Buccal cavity

- **Teeth** — chew and grind food (physical digestion), increasing surface area.
- **Tongue** — mixes food with saliva, helps in swallowing, has taste buds.
- **Salivary glands** secrete **saliva** containing enzyme **salivary amylase (ptyalin)**.
  - Starch → Maltose (sugar). Medium: slightly acidic/neutral.
- Food becomes a soft slippery **bolus**.

(b) Oesophagus (food pipe)

- Food is pushed down by rhythmic contraction of muscles — **peristaltic movement** (peristalsis). Occurs all along the gut.

(c) Stomach

- Thick muscular walls; churns food.
- **Gastric glands** secrete **gastric juice**:
  1. **Hydrochloric acid (HCl)** — creates acidic medium for pepsin to act; kills bacteria/germs.
  2. **Pepsin** — digests **proteins → peptones/peptides**.
  3. **Mucus** — protects the inner lining of the stomach from the action of acid.
- Exit of food regulated by the **pyloric sphincter** muscle.
- Semi-digested food is now called **chyme**.

> Acidity/ulcers: Excess acid → burning sensation; antacids (bases) neutralise it. Ulcers are often caused by the bacterium Helicobacter pylori.

(d) Small Intestine

- **Longest part** of the alimentary canal (~6 m); fitted into a compact space by coiling.
- Length varies with diet: **herbivores have a longer small intestine** (cellulose digestion takes time); **carnivores have a shorter one** (meat is easier to digest).
- Site of **complete digestion** of carbohydrates, proteins and fats.
- Receives secretions from **liver** and **pancreas**.

Bile juice (from liver, stored in gall bladder):

- **Alkaline** — makes the acidic food from the stomach alkaline for pancreatic enzymes to act.
- **Emulsification of fats** — breaks large fat globules into smaller globules, increasing efficiency of enzyme action. (Bile has no enzymes.)

Pancreatic juice (from pancreas):

- **Trypsin** → digests proteins

- **Pancreatic amylase** → digests starch
- **Lipase** → breaks down emulsified fats

Intestinal juice (from glands in the intestinal wall):

- Completes digestion:
  - Carbohydrates → **Glucose**
  - Proteins → **Amino acids**
  - Fats → **Fatty acids + Glycerol**

Absorption:

- Inner lining of small intestine has finger-like projections — **villi**.
- Villi **increase surface area** for absorption; richly supplied with blood vessels.
- Absorbed food is carried by blood to every cell for **energy, building new tissues and repair**.

(e) Large Intestine

- Absorbs **water** (and some salts) from the unabsorbed food.
- Remaining waste is **egested** through the **anus**, regulated by the **anal sphincter**.

Summary table of digestion:

Region	Secretion	Enzyme	Action
Mouth	Saliva	Salivary amylase	Starch → Maltose
Stomach	Gastric juice	Pepsin (+HCl, mucus)	Proteins → Peptones
Small intestine	Bile (liver)	–	Emulsifies fats, makes medium alkaline
Small intestine	Pancreatic juice	Trypsin, Amylase, Lipase	Proteins→amino acids; Starch→sugar; Fats→fatty acids+glycerol
Small intestine	Intestinal juice	Erepsin, maltase, lipase	Completes digestion

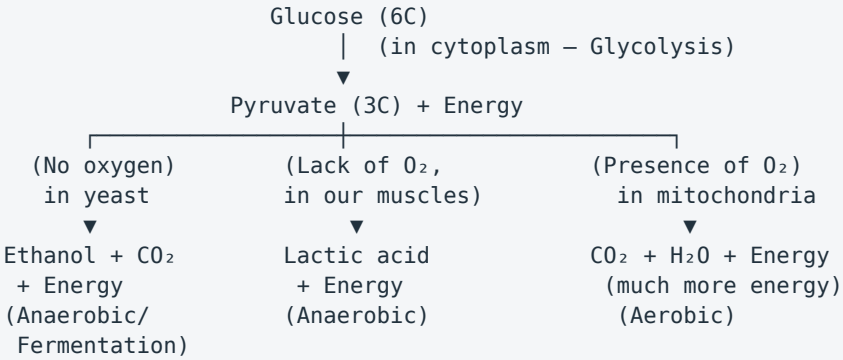
3. RESPIRATION

Respiration: The process of breaking down food (glucose) to release energy, which is stored as **ATP**.

Breathing vs Respiration:

- Breathing — physical process of inhaling O_2 and exhaling CO_2.
- Respiration — biochemical process of oxidation of food inside cells to release energy.

3.1 Breakdown of Glucose



- **Glycolysis** occurs in the **cytoplasm**; aerobic respiration occurs in the **mitochondria**.
- **Mitochondria = "powerhouse of the cell."**
- **Aerobic respiration releases much more energy than anaerobic respiration.**
- **Muscle cramps** occur during sudden vigorous activity due to accumulation of **lactic acid** (anaerobic respiration in muscles due to lack of oxygen).

ATP (Adenosine Triphosphate): The **energy currency** of the cell.

- ADP + Pi + energy → ATP
- Breakdown of one ATP molecule releases **30.5 kJ/mol** of energy.
- ATP is used for muscle contraction, protein synthesis, nerve impulse conduction, etc.

3.2 Respiration in Plants

- Exchange of gases through **stomata** (leaves) and **lenticels** (stems), and through root surfaces.
- Direction of gas exchange depends on need & time:
  - **Night:** only respiration → CO_2 released.
  - **Day:** photosynthesis dominates → net O_2 released; CO_2 produced in respiration is used up.

3.3 Respiration in Aquatic vs Terrestrial Organisms

- Water contains **dissolved oxygen** which is **far less** than the amount of O_2 in air.
- Hence aquatic animals **breathe faster** and use **gills** (large surface area, richly supplied with blood vessels).
- Terrestrial organisms use atmospheric O_2 — lungs, tracheae (insects), skin (earthworm), etc.

3.4 Human Respiratory System

Path of air: Nostrils → Nasal cavity → Pharynx → Larynx → Trachea (windpipe) → Bronchi → Bronchioles → Alveoli

- **Nostrils:** Hair and mucus filter dust; air is warmed and moistened.
- **Trachea:** Supported by **rings of cartilage** so that it does not collapse when there is less air.
- **Alveoli:** balloon-like structures — **site of gas exchange**.
  - Provide a very **large surface area** (~80 m² in humans).
  - Walls are **thin (one cell thick)** and contain an extensive network of **blood capillaries**.
  - O_2 diffuses into blood; CO_2 diffuses out.

Mechanism of breathing:

- **Inhalation:** Diaphragm flattens (moves down), ribs lift up and outward → chest cavity volume increases → air rushes in.

- **Exhalation:** Diaphragm moves up (domes), ribs move down and inward → volume decreases → air pushed out.
- **Residual volume:** Lungs always contain a residual volume of air so that there is sufficient time for oxygen to be absorbed and CO_2 to be released.

Transport of gases:

- **Oxygen** is carried by the respiratory pigment **haemoglobin** (in RBCs), which has a high affinity for oxygen. Its deficiency causes **anaemia**.
- **CO_2** is more soluble in water, so it is mostly transported in **dissolved form in the plasma**.

> **Rate of breathing:** ~15–18 times/minute in an adult at rest.

4. TRANSPORTATION

4.1 TRANSPORTATION IN HUMAN BEINGS

Blood = a **fluid connective tissue**. Components: **Plasma** (fluid medium) + **RBCs** + **WBCs** + **Platelets**.

- Plasma transports food, CO_2, nitrogenous wastes in dissolved form.
- RBCs carry oxygen (haemoglobin).
- Platelets help in **clotting** of blood at the site of injury, preventing loss of blood.

The Human Heart

- A **muscular organ**, about the size of our fist.
- Has **four chambers**: Right atrium, Right ventricle, Left atrium, Left ventricle.
- **Why four chambers?** To prevent the mixing of oxygenated and deoxygenated blood — this allows a **highly efficient supply of oxygen**, needed by warm-blooded animals (birds & mammals) to maintain body temperature.

Flow of blood (Double circulation):

1. **Deoxygenated** blood from body → **Vena cava** → **Right atrium** → (contracts) → **Right ventricle** → **Pulmonary artery** → **Lungs** (oxygenated)
2. **Oxygenated** blood → **Pulmonary vein** → **Left atrium** → (contracts) → **Left ventricle** → **Aorta** → **Whole body**

- **Valves** ensure that blood flows only in one direction (prevent backflow).
- **Left ventricle has the thickest walls** because it pumps blood to the entire body at high pressure.
- When atria contract, ventricles are relaxed, and vice versa.

Double Circulation: Blood passes through the heart **twice** in one complete cycle of the body.

- **Pulmonary circulation:** Heart → Lungs → Heart
- **Systemic circulation:** Heart → Body → Heart

> **Note:** Fishes have a **two-chambered heart**; blood goes only once through the heart (single circulation). Amphibians and many reptiles have **three-chambered hearts** and tolerate some mixing of blood since they do not regulate body temperature.

Blood Vessels

Feature	Arteries	Veins	Capillaries
Direction	Carry blood **away** from heart	Carry blood **to** the heart	Connect arteries & veins
Blood	Oxygenated (except pulmonary artery)	Deoxygenated (except pulmonary vein)	–
Walls	Thick, elastic (high pressure)	Thin	**One-cell thick**
Valves	Absent	**Present** (prevent backflow)	Absent

Blood Pressure:

- **Systolic pressure:** pressure in artery during ventricular contraction — normal **120 mm Hg**.
- **Diastolic pressure:** pressure during ventricular relaxation — normal **80 mm Hg**.
- Measured by a **sphygmomanometer**.
- **Hypertension (high BP):** 150/90 mm Hg or more — caused by narrowing of arterioles; can lead to rupture of arteries and internal bleeding.

Lymph (Tissue fluid)

- Formed when some plasma, proteins and blood cells escape through the pores of capillaries into intercellular spaces.
- Colourless, contains less protein than blood.
- **Functions:** Carries digested and **absorbed fats from the intestine**; drains **excess fluid** from extracellular space back into the blood; part of the immune system.

4.2 TRANSPORTATION IN PLANTS

Plants have low energy needs and use a slow transport system. Two conducting tissues:

(a) XYLEM — transport of water and minerals

- Made of **tracheids and vessels** (dead cells) — form continuous tubes from root to leaf.
- **Mechanism:**
  1. **Root pressure:** Ions are actively taken up by root cells → increases ion concentration → water moves in from soil by **osmosis** → creates a column of water pushed upward. (Significant at night.)
  2. **Transpiration pull:** Evaporation of water from stomata creates a suction that pulls water up from the xylem. This is the **main driving force in daytime**, especially in tall plants.

Transpiration: Loss of water in the form of vapour from the aerial parts of a plant. **Functions of transpiration:**

- Helps in the absorption and upward movement of water and minerals.
- Helps in **temperature regulation** (cooling) of the plant.

(b) PHLOEM — translocation of food

- **Translocation:** Transport of soluble products of photosynthesis (mainly **sucrose**), amino acids and other substances.
- Made of **sieve tubes and companion cells** (living).
- Transport is **bidirectional** — from leaves to storage organs (roots, fruits, seeds) and from storage organs to growing parts.

- **Mechanism:** Material is transferred into sieve tubes using **energy from ATP**; this increases osmotic pressure → water enters → pressure moves the material to tissues of lower pressure.

Xylem vs Phloem:

Xylem	Phloem
Transports water & minerals	Transports food (sucrose, amino acids)
Unidirectional (upward)	Bidirectional
Dead cells (mostly)	Living cells
Uses physical forces (transpiration pull) — no energy	Requires energy (ATP)

5. EXCRETION

Excretion: The biological process of removing harmful **nitrogenous metabolic wastes** from the body.

5.1 Human Excretory System

Organs: A pair of **kidneys**, a pair of **ureters**, a **urinary bladder**, and a **urethra**.

- **Kidneys:** Located in the abdomen, one on either side of the backbone. Filter blood and form urine.
- **Ureters:** Carry urine from kidneys to urinary bladder.
- **Urinary bladder:** Stores urine (muscular, under nervous control).
- **Urethra:** Urine passes out through it.

Main excretory product: Urea (formed in the **liver** from the breakdown of excess amino acids). Others: uric acid, creatinine, excess water, salts.

The Nephron

- **Nephron = basic filtration unit of the kidney.** Each kidney has about **one million nephrons**.

Parts & Steps:

1. **Glomerulus** — a cluster of thin-walled capillaries inside **Bowman's capsule**. Blood enters through the **renal artery** under high pressure → **ultrafiltration**: water, glucose, amino acids, salts, urea pass into the Bowman's capsule. Blood cells and proteins are retained.
2. **Tubular part (selective reabsorption)** — as the filtrate flows along the tubule, useful substances such as **glucose, amino acids, salts and most water** are **selectively reabsorbed** into the blood capillaries surrounding the tubule.
3. **Collecting duct** — collects urine from many nephrons → carried to ureter.

Amount of water reabsorbed depends on:

- How much **excess water** there is in the body.
- How much **dissolved waste** needs to be excreted.

> Humans produce about **1–1.8 litres of urine per day**.

Artificial Kidney (Haemodialysis)

- Used when kidneys fail (kidney failure → toxins accumulate → death).
- A machine containing tubes made of **selectively permeable membrane** placed in a tank of **dialysing fluid** (has the same osmotic pressure as blood but **no nitrogenous wastes**).
- Waste products diffuse out from the blood into the dialysing fluid; purified blood is returned to the body.

- **Difference from kidney:** There is **no reabsorption** in an artificial kidney.

5.2 Excretion in Plants

Plants lack a special excretory organ. They use several strategies:

- **Oxygen and CO_2** — released through **stomata and lenticels**.
- **Excess water** — removed by **transpiration**.
- Waste products stored in **cell vacuoles**, or as **resins and gums** especially in **old xylem**.
- Some wastes stored in **leaves that fall off**.
- Some wastes excreted into the **soil** around them.

6. QUICK REVISION — IMPORTANT POINTS

- Chlorophyll → in chloroplast; "kitchen of the cell."
- Mitochondria → "powerhouse of the cell"; ATP → "energy currency."
- Guard cells regulate the opening/closing of stomata.
- Herbivores → longer small intestine; carnivores → shorter.
- Bile: no enzyme, but emulsifies fats & makes medium alkaline.
- Villi → increase surface area for absorption.
- Alveoli → increase surface area for gas exchange.
- Haemoglobin → transports O_2; CO_2 transported dissolved in plasma.
- Four-chambered heart → prevents mixing of oxygenated & deoxygenated blood.
- Double circulation → blood passes through heart twice per cycle.
- Xylem → water (upward, transpiration pull); Phloem → food (bidirectional, ATP).
- Nephron → structural & functional unit of kidney.
- Urea → made in liver, removed by kidney.

7. COMMON BOARD QUESTIONS

1. Why is diffusion insufficient to meet the oxygen requirements of multicellular organisms like humans?
2. Describe the process of nutrition in Amoeba with a diagram.
3. What are the different ways in which glucose is oxidised to provide energy in various organisms?
4. How are the lungs designed in human beings to maximise the area for exchange of gases?
5. What are the components of the transport system in human beings? What are the functions of these components?
6. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?
7. How is water and minerals transported in plants? / How is food transported in plants?
8. Describe the structure and functioning of nephrons.
9. What are the methods used by plants to get rid of excretory products?
10. How is the amount of urine produced regulated?

Diagrams to practise:

- Cross-section of a leaf / Stomatal apparatus
- Human digestive system
- Human respiratory system & alveoli
- Human heart (with labelled blood flow)
- Excretory system & structure of a nephron